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Project Title	Zomato Data Analysis
Technologies	Data Analytics using SQL
Domain	Online Food Supply
Project Difficulty level	Beginner

Problem Statement

The primary objective of this analysis is to obtain a comprehensive overview of the dynamics within Zomato's business ecosystem. By leveraging SQL querying and analysis techniques, we aim to uncover valuable insights that shed light on various aspects of the company's functioning.

About Zomato

Zomato, a prominent Indian multinational restaurant aggregator and food delivery company, plays a significant role in the food service industry. To gain deeper insights into its operations, we intend to conduct an exploratory analysis of the Zomato dataset. This dataset comprises over 9000 rows and encompasses crucial information such as Restaurant ID, Restaurant Name, City, Location, Cuisines, and more.

Dataset Description:

Their columns include the following:

- **RestaurantID:** A unique identifier assigned to each restaurant in the dataset.
- **RestaurantName:** The name of the restaurant, providing its distinctive identity.
- **CountryCode:** The numerical code representing the country where the restaurant is located.
- **City:** The city in which the restaurant is situated.
- **Address:** The specific address where the restaurant is physically located.
- **Locality:** A specific neighborhood or area within the city where the restaurant is situated.
- **LocalityVerbose:** A verbose description of the locality, offering additional context.
- **Cuisines:** The variety of cuisines offered by the restaurant, highlighting its culinary specialties.

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- **Currency:** The currency used for pricing within the restaurant's menu.
- **Has_Table_booking:** A binary indicator (0 or 1) signifying whether the restaurant offers table booking services.
- **Has_Online_delivery:** A binary indicator (0 or 1) representing whether the restaurant provides online delivery services.
- **Is_delivering_now:** A binary indicator (0 or 1) indicating if the restaurant is currently delivering orders.
- **Switch_to_order_menu:** A binary indicator (0 or 1) suggesting whether customers need to switch to an order menu.
- **Price_range:** A categorical variable indicating the price range of the restaurant's offerings.
- Votes: The total number of votes or ratings received by the restaurant.
- **Average_Cost_for_two:** The approximate cost for two people dining at the restaurant.
- **Rating:** The average rating assigned to the restaurant based on customer reviews and ratings.

Tasks:

Data Exploration:

Check Table Details:

- Review the table structure including column names, data types, and constraints.

Duplicate Check:

- Investigate duplicate values in the [RestaurantId] column to ensure data integrity.

Column Removal:

- Identify and remove unnecessary columns to streamline the dataset.

Table Merging with New Column:

- Merge two distinct tables and create a new column [Country_Name] using [CountryCode] as a primary key.

Correcting Misspelled City Names:

- Detect and correct any mis-spelled city names for consistency and accuracy.

Rolling Count Analysis:

- Use window functions for rolling count/moving count to determine the number of restaurants.

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Statistical Summary:

- Calculate statistical measures (min, max, avg) for votes, rating, and currency columns.

New Category Creation:

- Create a new category column based on specific criteria for ratings.

Data Analysis:

Country Distribution Analysis:

- Calculate the percentage of data related to restaurants in each country.

Online Delivery Insights:

- Determine the percentage of restaurants offering online delivery in different countries.
- Identify the countries where online delivery is most prevalent.

Focus on Indian Restaurants:

- Analyze data specific to Indian restaurants for deeper insights.

City Analysis - Popular Locations:

- Identify cities with the highest number of listed restaurants.
- Highlight the top three cities with the most listed restaurants.

Cuisine Insights:

- Determine the most popular cuisines in a specific city (Connaught Place).

Table Booking Impact Analysis:

- Compare the number of restaurants with table booking facility to those without in Connaught Place.
- Explore the average ratings for both categories.

Restaurant Recommendations:

- Identify best moderately priced restaurants meeting specific criteria (average cost, rating, votes, facilities).
- Locate the best restaurant meeting these criteria in a specific city (Kolkata).

Dataset:

You can find the dataset on the Attached mail

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Approaches

Feel free to leverage the latest capabilities of MySQL and employ a diverse range of techniques that resonate with your innovative mindset. Your imaginative approach to crafting solutions is not only welcomed but also valued and encouraged.

Project Evaluation Metrics

- **Design and Construct:** Build the MySQL system adhering to the provided requisites.
- **Modular Structure:** Arrange the system components using a modular structure to enhance comprehensibility.
- **User-Centric:** Guarantee that the system prioritizes user convenience and security, minimizing potential hazards.
- **Test-Ready:** Develop the system components in a manner that facilitates robust testing procedures.
- **Sustainability:** Architect the system to enable straightforward modifications and improvements when necessary.
- **Universal Compatibility:** Certify that the system performs consistently across various MySQL environments and editions.

Submission requirements:

Project Task:

Ensure that your MySQL system comprehensively encompasses the stipulated visualizations and analysis.

MySQL: You are expected to furnish the MySQL system, fully refined and operational, in its conclusive configuration.

Detail project report:

You have to create a detailed project report and submit that document as per the given

Sample

https://docs.google.com/presentation/d/1_fMRxgL4eEPWIdW2eqwCxejVUtCVOxtPoy3At_AKo5I/edit?usp=sharing

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